

```

// BASE CASE
if (permLength === 0) { ❶
    document.write(indent + "Base case reached, returning " + [prefix] + "<br />");
    return [prefix];
}

// RECURSIVE CASE
// Create a new prefix by adding each character to the current prefix.
let results = [];
document.write(indent + "Adding each char to prefix \"" + prefix + "\".<br />");
for (char of chars) {
    let newPrefix = prefix + char; ❷

    // Decrease permLength by one because we added one character to the prefix.
    results = results.concat(getPermsWithRep(chars, permLength - 1, newPrefix)); ❸
}
document.write(indent + "Returning " + results + "<br />");
return results;
}

document.write("<pre>All permutations with repetition of JPB123:<br />");
document.write(getPermsWithRep('JPB123', 4) + "</pre>");
</script>

```

The output of these programs is shown here:

```

All permutations with repetition of JPB123:
Start, args=("JPB123", 4, "")
Adding each char to prefix "".
.Start, args=("JPB123", 3, "J")
.Adding each char to prefix "J".
..Start, args=("JPB123", 2, "JJ")
..Adding each char to prefix "JJ".
...Start, args=("JPB123", 1, "JJJ")
...Adding each char to prefix "JJJ".
....Start, args=("JPB123", 0, "JJJJ")
....Base case reached, returning ['JJJJ']
....Start, args=("JPB123", 0, "JJJP")
....Base case reached, returning ['JJJP']

```